

Basil K Paul

Full Stack Python Developer — Backend, APIs & ML Integration

+91-8943553166 basilpaul71136@gmail.com linkedin.com/in/basil-k-paul-6599a0249 github.com/n8novaa

Summary

Computer Science graduate student with hands-on experience building full-stack Python applications across two industry internships and four shipped projects. Strong backend foundation in Django, Django REST Framework, OOP design, and relational databases; comfortable integrating AI/ML workflows using Scikit-learn and Pandas into web applications. Experienced building responsive frontends with React and JavaScript. Eager to contribute to a product-focused team working on real-world software solutions.

Skills

Backend: Python, Django, Django REST Framework, OOP, REST API Design, JWT Authentication

AI / Machine Learning: Scikit-learn, Pandas, NumPy, Feature Engineering, Supervised Learning

Databases: SQL, Relational Database Design, Firebase Realtime Database

Frontend: React.js, HTML, Tailwind CSS, JavaScript, Responsive Design

Tools: Git, GitHub, Postman

Other: Kotlin, Android SDK, ESP32, IoT Sensor Integration

Experience

- **Machine Learning Intern** June 2025 – July 2025
NeST Cyber Campus, Kochi
 - Built end-to-end Python data pipelines on structured industry datasets: handled missing values, engineered features, and normalised inputs to improve data quality.
 - Trained and evaluated supervised classification and regression models using Scikit-learn; applied cross-validation to ensure reliable performance estimates.
 - Packaged model outputs into reusable Python prediction pipelines, enabling seamless integration into downstream application workflows.
- **Python Full Stack Developer Intern** June 2024 – July 2024
Futuro IT Solutions
 - Developed and shipped full-stack features using Python and Django for a production web application, covering backend logic, database interactions, and UI integration.
 - Built, tested, and documented RESTful API endpoints (JSON) integrated with relational databases; debugged and maintained the existing codebase.
 - Reduced code duplication through modular OOP-based backend component design, improving long-term maintainability.

Projects

- **Book Recommendation System | Python, Django, Scikit-learn, React** 2026
github.com/n8novaa/book_recommendation_advanced
 - Built a hybrid AI recommendation engine combining matrix factorization, TF-IDF cosine similarity, and genre preference weighting, served through a Django REST API backend.
 - Implemented cold-start fallback, diversity control, and a caching layer with session audit logging to handle edge cases and improve recommendation quality.
 - Built a fully responsive React frontend consuming the REST API, delivering AI-powered personalised recommendations based on each user's interaction history.

• **Studygram | Python, Django REST Framework, React | github.com/n8novaa/studygram 2025**

- Built a Discord-style collaborative study platform with real-time chatrooms, topic-based discussion threads, and full CRUD functionality throughout.
- Designed the backend using Django REST Framework with a modular service architecture, enabling independent development and extension of features.
- Implemented clean API contracts between the frontend and backend, ensuring a consistent and scalable data flow across the application.

• **PortfolioHub | Python, Django, SQL | github.com/n8novaa/portfoliohub 2025**

- Developed a multi-user portfolio platform with JWT authentication, supporting dynamic profile and project management per user.
- Designed a normalised SQL schema and REST API layer that cleanly separates data access from business logic across the application.
- Applied OOP principles throughout the Django backend for a clean, reusable, and maintainable codebase.

• **Novacane | IoT, Kotlin, Firebase, ESP32 | github.com/n8novaa/NovaCane 2026**

- Built a smart assistive cane for visually impaired users using ESP32 with ultrasonic and proximity sensors for real-time obstacle detection; selected as B.Tech final-year project and **publicly demonstrated at a college expo**.
- Integrated sensor firmware, BLE/Wi-Fi communication, and a Firebase Realtime Database backend into a unified end-to-end system achieving sub-second alert latency.
- Developed a dual-interface Android app in Kotlin for both the user and a remote guardian, with real-time location tracking and emergency alerts; work led to a peer-reviewed IJIRT publication.

Research & Publication

- **Published** in International Journal of Innovative Research and Technology (IJIRT): *A Survey on AI-Powered Smart Assistive Cane for Visually Impaired Using IoT and Computer Vision* — peer-reviewed research underpinning the Novacane project.

Education

• **B.Tech in Computer Science and Engineering** 2022 – 2026
Sree Narayana Gurukulam College of Engineering, Kochi, Kerala CGPA: 7.0 / 10

Relevant Coursework

Data Structures and Algorithms | Database Management Systems | Operating Systems | Machine Learning | Computer Networks

Leadership & Activities

- Coordinator of the college dance team; organised inter-college cultural events. Volunteer, National Service Scheme (NSS).